

Benjamin A. Batya

<https://www.linkedin.com/in/benjaminbatya/>

benjaminbatya@gmail.com

<https://github.com/benbatya>

SUMMARY OF QUALIFICATIONS

A seasoned Senior Software Engineer with a career spanning more than two decades, specialized in the architecture and optimization of high-performance systems. Expertise includes AI development techniques, embedded systems, user interface/user experience design, additive manufacturing, and extended reality (XR) platforms. Recognized for directing cross-functional teams to execute comprehensive, full-lifecycle software solutions across a multitude of operating systems and environments. Offers a versatile and extensive technical background applicable to various project scales and development team structures.

TECHNICAL SKILLS

- **Soft Skills:** UX/UI Design, Mentorship, Roadmapping, XFN Collaboration
- **Languages:** C/C++(99-23), Typescript, React, Rust, Python, C#, Kotlin, Java, Hack, SQL
- **UI/Graphics:** Vulkan, OpenGL, WebGL, React, ImGui, Qt/QML, VTK
- **Design Tools:** Unity, Unreal Engine, Godot, Ronin, Maya, Saleae
- **Platforms/OS:** Android, Windows, MacOS, Linux, QNX, Xbox360, PS3, nRF52
- **Build/Source:** Buck, Skylark, Soong, Make, CMake, Mercurial, Git, Perforce
- **Schema Formats:** GRPC, Avro, XML, JSON-Schema, YAML

EMPLOYMENT HISTORY

Generation Alpha Transistor - Technical Consultant

April 2026 - July 2026

- Architected test infrastructure for Pull Requests and AI Evaluations, reducing durations by 90%
- Redesigned the EDA-IC UI to provide domain-specific functionality and enhanced experience

Meta Platforms - Senior Software Engineer

February 2021 - January 2026

- Led development on the **Meta XR Simulator** (XrSim) which reduced iteration time for headset projects by up to **50%** for almost **9,000** MAU
- Designed synthetic environment infrastructure for fast iteration of **Mixed Reality** applications
- Ensured **Multiplayer** test scenarios worked without resource contention
- Built **OpenXR** integrations for **Unity & Unreal** and supported **Godot & Android Studio**
- Guaranteed performance and reliability for **Linux, Android, MacOS, and Windows**
- Architected integrated rendering capabilities across **Metal, Vulkan, DX11, and DX12** frameworks
- Mentored 8+ junior engineers through design and code reviews, fostering professional growth and project impact
- Optimized cross-functional efficiency by implementing robust design workflows and test infra
- Maintained build and test infrastructure in **Buck, Skylark, and Hack**
- Ported internal emulator stack based on mobile OS to X86_64 Linux

Tensil.io - Senior Software Engineer

June 2020 - November 2020

- Developed an in-browser multi-person video editor using **Wasm** (C++ & TS)
- Implemented **frame-accurate** playback and scrubbing infrastructure
- Improved clip load times by **50%** by prefetching and caching frames

Vytronus - Senior Software Engineer

September 2018 - January 2020

- Improved rendering of **ultrasonic catheter** and ring sensors within heart (C++)
- Enhanced position accuracy using the **Biarc algorithm** to achieve **1mm** precision
- Developed real-time UI components for **surgery system** with VTK and QML

Stealth Neural Sim Startup - Software Consultant

May 2018 - September 2018

- Integrated **nRF52 SoC** with Android app to control bandaid-sized neural sim
- Designed embedded control software for voltage control and **BLE interface**
- Tested system with **Saleae** Logic Analyser

Velo3D - Software Engineer

July 2016 - March 2018

- Engineered a pre-print system featuring CAD import, part placement, and support generation.
- Designed a **hierarchical spreadsheet** for developing additive manufacturing processes.
- Reduced data entry errors through domain constraints and structured data entry implementation.
- Supported successful development of the final **Inconel**, **Stainless Steel**, and **Ti** processes.

Holo3D.tv - Co-Founder and Engineer

September 2014 - April 2016

- Designed a prototype for an **interactive sports broadcasting** platform.
- Built a prototype in **Unreal 4** and pitched it to multiple VCs
- Developed a point cloud capture&viewing system using **Kinect** cameras (C++/Qt)

Zynga - Software Engineer

December 2009 - June 2010

- Implemented asset loader for **Cafeworld** that decreased loading time by **80%**
- Developed **Actionscript** gameplay mechanics to improve MAU
- Managed development server infrastructure to support gameplay engineering.

LucasArts - Tools Engineer

August 2007 - December 2009

- Developed tools and workflows for **Star Wars: TFU** and **TFU2**, optimizing console-specific development pipelines.
- Improved iterative desktop->console test time from minutes to seconds.
- Built level design and remoting tools for gameplay engineers and designers.
- Generated schemas from C++ classes to accelerate development of gameplay systems.

Versamed - UI Engineer

July 2005 - July 2007

- Built GUI for the **iVent101** medical ventilator, enhancing usability for doctors and patients.

EDUCATION

University of California, Berkeley, CA

September 1997 - May 2001

- Bachelors of Engineering in Electrical Engineering & Computer Science program